Azure Security Engineering Lab 01 — Identity & Access Control

Project: Azure Security Foundation & Identity Access Control
Platform: Microsoft Azure
Technologies: Microsoft Entra ID, Azure RBAC, Azure Resource Manager, Azure Cost Management
Project Type: Hands-on Cloud Security Lab
Certification Preparation: Microsoft SC-500

Project Overview

I built a hands-on Azure security lab to develop practical experience in identity and access management (IAM), Azure role-based access control (RBAC), and the principle of least privilege.

The objective was to create a controlled Azure environment, configure test identities and security groups, assign permissions, and validate access at the resource-group level.

The lab focused on understanding how identity management and resource authorization work together to secure cloud environments.

Lab Objectives

- Establish a dedicated Azure resource group for security testing.
- Explore the relationship between an Azure subscription and a Microsoft Entra tenant.
- Create and manage cloud-based test users.
- Organize identities using Microsoft Entra security groups.
- Configure Azure RBAC roles at a defined scope.
- Validate effective access and investigate inherited permissions.
- Apply least-privilege security principles.
- Review resource costs and plan secure lab cleanup.

Lab Architecture

The environment consisted of one Microsoft Entra tenant, one Azure subscription, and a dedicated resource group named rg-sc500-lab.

Two test users were created:

User	Security Group	Azure RBAC Role
Alex Morgan	SG-SC500-Readers	Reader
Jordan Lee	SG-SC500-Operators	Contributor

Both roles were assigned at the resource-group scope rather than across the subscription.

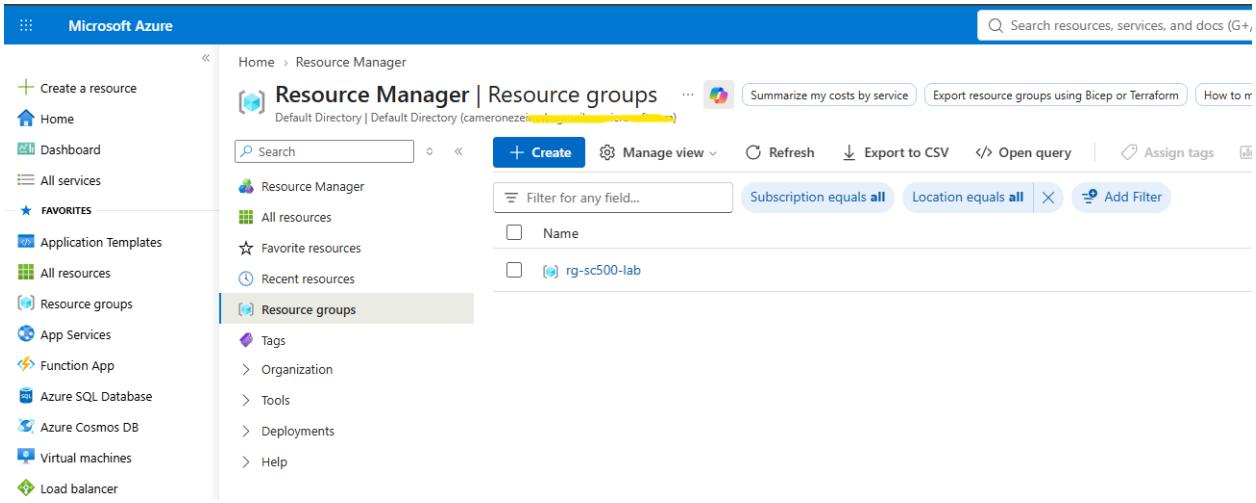
Implementation

Phase 1 — Azure Environment Setup

I accessed the Azure Portal and explored my Azure subscription and Microsoft Entra tenant.

I created a dedicated resource group named **rg-sc500-lab** to provide an isolated scope for the lab's Azure RBAC configurations.

Cameron Ezeiruaku



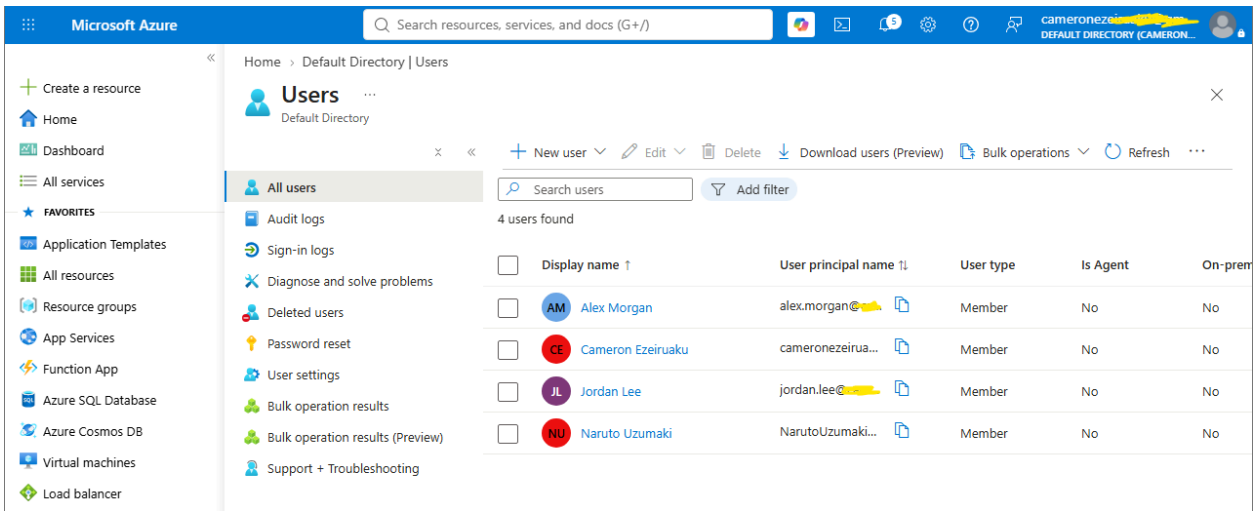
I also reviewed Azure Cost Management and the process for establishing a monthly budget and spending notifications.

Key concepts: Subscription scope, resource groups, Entra tenants, cost governance.

Phase 2 — Identity and Group Management

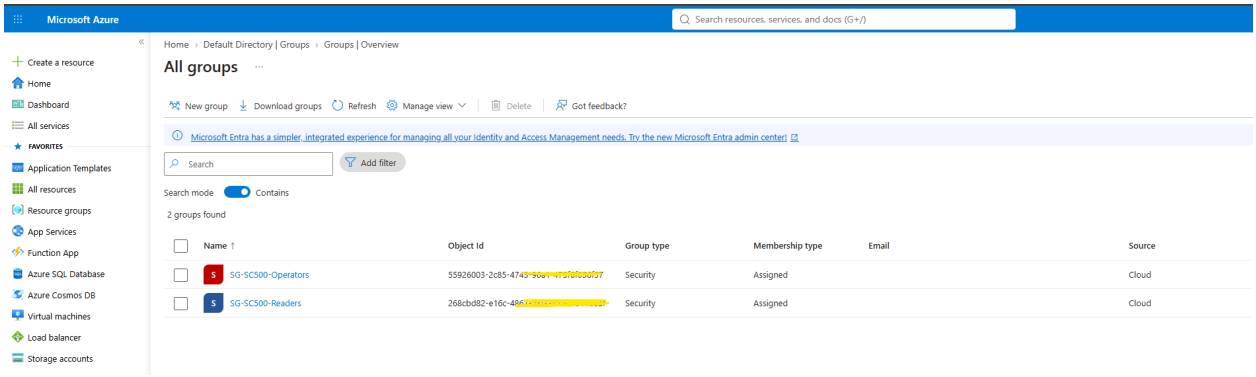
I created two cloud-based test identities in Microsoft Entra ID: Alex Morgan and Jordan Lee.

Note: (Naruto Uzumaki was the first test account I made outside of this lab)



I then created two security groups:

- **SG-SC500-Readers**
- **SG-SC500-Operators**



Alex Morgan was added to the Readers group, and Jordan Lee was added to the Operators group.

This established a group-based access management structure, allowing permissions to be assigned to groups rather than individual users.

Key concepts: Cloud identities, group membership, centralized access management.

Phase 3 — Azure RBAC Configuration

I configured two Azure RBAC role assignments using the resource group's Access Control (IAM) interface.

Reader: Assigned to **SG-SC500-Readers**, providing read-only resource management permissions.

Contributor: Assigned to **SG-SC500-Operators**, permitting management of most Azure resources while excluding Azure RBAC role-assignment management.

Both role assignments were scoped specifically to **rg-sc500-lab**.

This design demonstrated how granular scope selection can reduce unnecessary access across a subscription.

Key concepts: Security principals, built-in roles, role assignments, least privilege.

Phase 4 — Access Validation

I reviewed effective access for both test identities using Azure's Access Control (IAM) interface.

The validation exercises included:

- Verifying Alex Morgan's Reader role through group membership.
- Verifying Jordan Lee's Contributor role through group membership.
- Examining the management actions permitted by each role.
- Reviewing why Contributor cannot assign Azure RBAC roles without additional privileges.
- Inspecting the inheritance of role assignments through Azure's scope hierarchy.
- Reviewing how additional inherited permissions could affect effective access.

These exercises helped demonstrate the importance of validating actual user access rather than relying solely on intended role assignments.

Key concepts: Effective permissions, role inheritance, authorization, privilege boundaries.

Phase 5 — Security Cleanup

The cleanup procedure was designed to preserve reusable test identities and groups while eliminating unnecessary access.

The planned cleanup includes:

- Removing temporary Reader and Contributor role assignments.
- Disabling unused test accounts.
- Retaining the reusable Entra security groups.
- Reviewing the resource group for unwanted resources.
- Inspecting Azure Cost Management for unexpected charges.

Security Concepts Demonstrated

Principle of Least Privilege: Assigning only the permissions necessary to perform an intended function.

Role-Based Access Control: Managing authorization through defined roles rather than granting broad administrative permissions.

Group-Based Access Management: Using security groups to simplify user onboarding, permission changes, and removal of access.

Scope Inheritance: Understanding how permissions assigned at higher Azure scopes flow to child resources.

Separation of Duties: Assigning different levels of access based on responsibilities.

Access Lifecycle Management: Reviewing and removing temporary privileges after testing is complete.

Lessons Learned

This lab reinforced several important Azure security concepts.

First, an Entra user or group does not automatically have Azure resource access. An Azure RBAC assignment must authorize the appropriate principal at the required scope.

Second, selecting the correct role is only one part of access security. The assignment scope is equally important.

Third, effective permissions may include multiple assignments, including permissions inherited from parent scopes.

Finally, temporary security privileges should be reviewed and removed after they are no longer required.

Tools and Technologies

- Microsoft Azure Portal
- Microsoft Entra ID
- Azure Role-Based Access Control (RBAC)
- Azure Resource Manager
- Azure Cost Management

Future Improvements

This lab establishes a foundation for more advanced Azure security projects.

Planned extensions include:

- Microsoft Entra Conditional Access and multifactor authentication
- Azure managed identities
- Azure Key Vault access control
- Azure Policy and governance
- Microsoft Defender for Cloud
- Security monitoring and audit logs

References

- Microsoft Entra ID documentation
- Azure RBAC overview
- Assign Azure roles using the Azure Portal
- Check access to Azure resources
- Azure Cost Management documentation

Project Outcome: Established and tested foundational Azure identity and access-control configurations using Entra ID, security groups, Azure RBAC, and resource-group-scoped permissions.